THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS FULL SYLLABUS PAPER SET 09

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. $M(3,\ 2)$ is the midpoint of AB and $A(1,\ -4)$. The coordinates of B are: 1

(a) $(2,\ 6)$ **(b)** $(5,\ 8)$ **(c)** $(5,\ -8)$ **(d)** $(4,\ -2)$

2. The point $(7,\ 15)$ ___ the line $y = 2x + 1$. 1

(a) is the origin **(b)** lies on the line

(c) is the y-intercept of the line **(d)** does not lie on the line

3. A rational number in lowest terms has denominator $2^2 \times 5^4$. Its decimal expansion terminates after how many places? 1

(a) 5 **(b)** 6 **(c)** 3 **(d)** 4

4. The number $0.1\overline{6}$ in the form $\frac{p}{q}$ is: 1

(a) 6 **(b)** $\frac{1}{3}$ **(c)** $\frac{4}{15}$ **(d)** $\frac{1}{6}$

5. Which of the following is an identity? 1

(a) $(x+y)^2 = x^2 + 2xy + y^2$ **(b)** $x^2 - 1 = 24$ **(c)** $2x = 10$ **(d)** $x + 5 = 9$

6. An arc subtends an angle of 150° at the centre. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 75° **(b)** 105° **(c)** 150° **(d)** 300°

7. By Heron's formula, the area of a triangle with sides 8 cm, 15 cm and 17 cm is: 1

(a) 66 cm^2 **(b)** 60 cm^2 **(c)** 120 cm^2 **(d)** 54 cm^2

8. A fair die is rolled once. The probability of getting a factor of 12 is: 1

(a) $\frac{5}{7}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{2}{3}$

9. The sequence $1, 3, 6, 10, 15, \ldots$ (triangular numbers) is: 1

(a) neither an AP nor a GP **(b)** an AP with d = 5

(c) a GP with r = 3 **(d)** an AP with d = 2

10. Fermat claimed that every number of the form $2^{2^n} + 1$ is prime. Euler disproved this by showing that for $n = 5$ the number is: 1

(a) negative **(b)** a perfect square **(c)** prime **(d)** composite

11. A group of 15 students has an average of 82 marks and another group of 35 students has an average of 74 marks. The average of all 50 students is: 1

(a) 75.4 **(b)** 78.4 **(c)** 77.4 **(d)** 76.4

12. 250 mL of a solution with 20% salt is mixed with 750 mL of a solution with 12% salt. The concentration of salt in the mixture is: 1

(a) 16% **(b)** 14% **(c)** 15% **(d)** 13%

13. By Euclid's algorithm, gcd(84, 60) is: 1

(a) 13 **(b)** 24 **(c)** 6 **(d)** 12

14. The list of common divisors of 375 and 825, in increasing order, ends with: 1

(a) 75 **(b)** 188 **(c)** 76 **(d)** 825

15. The number of times Step 3 of Euclid's subtraction algorithm is applied while computing gcd(21, 6) is: 1

(a) 4 **(b)** 5 **(c)** 7 **(d)** 6

16. If the diagonals of a quadrilateral bisect each other, then it is a: 1

(a) rectangle **(b)** kite **(c)** trapezium **(d)** parallelogram

17. The diagonals of a convex quadrilateral: 1

(a) are always perpendicular **(b)** never intersect

(c) intersect each other **(d)** are always equal

18. **Assertion (A):** The pair x + y = 4 and 2x + 2y = 8 has infinitely many solutions. 1
Reason (R): Every pair of linear equations in two variables has either exactly one solution or infinitely many solutions.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

19. The pair of equations 3x + 2y = 7 and 6x + 4y = 9 has: 1

(a) a unique solution **(b)** exactly two solutions

(c) infinitely many solutions **(d)** no solution

20. Taking $\pi = \frac{22}{7}$, the volume of a cylinder of radius 3.5 cm and height 8 cm is: 1

(a) 616 cm^3 **(b)** 308 cm^3 **(c)** 154 cm^3 **(d)** 176 cm^3

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Check whether $P(1, 0)$ is equidistant from $A(2, 3)$ and $B(4, -1)$. 2

22. State the slope and y-intercept of $y = -4x + 9$ and say whether it shows linear growth or decay. 2

23. Side BC of cyclic quadrilateral ABCD is produced to E and $\angle DCE = 70°$. Find $\angle A$. 2

24. How many black triangles are there at Stages 4 and 5 of the Sierpiński triangle? 2

25. A student says “If a number is divisible by 6, then it is divisible by 2 and 3.” Write the converse and state whether both are true. 2

26. A cafe rates food, service and ambience with weights 4 : 3 : 2. A customer gives 5, 4 and 2. Find the weighted rating. 2

27. Without solving, say how many solutions $2x + 3y = 9$ and $4x + 6y = 18$ have. Give a reason. 2

28. If the edge of a cube is doubled, by what factor do its surface area and volume change? 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 40–60 words each.

29. **Case study:** A delivery company packs 144 apples and 108 oranges into identical boxes with no fruit left over. 3

(i) Which gcd gives the largest number of fruits possible in each box if boxes hold only one fruit type?

(ii) Compute it by the division algorithm.

(iii) How many boxes of each fruit?

30. **Case study:** A floor is tiled with copies of the irregular quadrilateral SOME with angles 70°, 80°, 100°, 110°. 3

(i) Find the sum of its angles.

(ii) Four copies meet at a point using each angle once. What is the total angle there?

(iii) Does this show that the plane can be tiled with it?

31. Given $y = 2x + 1$, complete the table: $x = 1, 2, 5, 7, 9, 12, 20$ and check which of the points lies on the line: $(4, 9)$ or $(6, 14)$. 3

32. Fill the blanks: $(px + a)(qx + b) = (__)x^2 + (__)x + __$. Verify with $(2x + 3)(3x + 1)$. 3

33. Show that $R(-5, -1)$, $B(-2, -5)$ and $C(4, -12)$ are not collinear. 3

34. Show that the lines $y = mx + d_1$ and $y = mx + d_2$ with $d_1 \neq d_2$ are parallel. 3

35. Prove that $5 - \sqrt{3}$ is irrational, given that $\sqrt{3}$ is irrational. 3

OR

Verify the distributive law $\frac{1}{2} \times \left(\frac{1}{4} + \frac{1}{3}\right) = \frac{1}{2} \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{3}$. 3

36. Show that the points $(1, 3), (2, 6), (3, 12)$ from the GP $3, 6, 12, \ldots$ do not lie on a straight line. 3

OR

Show that the sum of two consecutive triangular numbers is a perfect square; verify for t_{10} and t_9. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. A zoo has 50 deer with average weight 40 kg; males average 46 kg and females 36 kg. 5

(i) Form the equation for the number of males x.

(ii) Find the number of males and females.

(iii) A female deer of 42 kg joins the zoo. Find the new average weight of the females.

OR

Case study: Alloy P weighs 100 kg with 40% silver, Alloy Q weighs 150 kg with 20% silver, and Alloy R has 60% silver. When all three are melted together, the new alloy has 40% silver. 5

(i) Write the equation for the weight y of Alloy R.

(ii) Solve it.

(iii) Verify by finding the silver in the final alloy.

38. **Case study:** A school wants to seat 84 boys and 126 girls in rows so that each row has only boys or only girls and every row has the same number of students. **5**

(i) What is the largest row size?

(ii) Show the divisor lists' common part.

(iii) How many rows of boys and girls result?

(iv) Which algorithm would you use for 8400 and 12600 and why?

OR

Case study: Neha writes the algorithm: 'Start with most-recent-common-divisor = 1. For each k from 2 to min(m, n): if k divides both m and n, update to k.' **5**

(i) Execute it for m = 20, n = 30 and list the k that update the value.

(ii) What is the final value and what does it equal?

(iii) How many values of k were examined?

(iv) Suggest how to reduce the number of checks using Euclid's idea.

39. **Case study:** A rectangular garden $OABC$ has $O(0,0)$, $A(12,0)$, $B(12,8)$, $C(0,8)$. A fountain is at the centre F, and lamps $L_1(0,4)$, $L_2(12,4)$. **5**

(i) Find the centre.

(ii) Show $FL_1 = FL_2$.

(iii) Find the area of the garden and the distance between O and B.

OR

The vertices of $\triangle ABC$ are $A(-2,\ 0)$, $B(2,\ 0)$ and $C(0,\ 2)$, and those of $\triangle PQR$ are $P(-4,0)$, $Q(4,0)$, $R(0,4)$. Show that $\triangle ABC \sim \triangle PQR$ by finding the ratios of the sides. **5**

40. **Case study:** Mādhava's series gives $\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right)$ and Āryabhaṭa gave $\pi \approx 3.1416$. **5**

(i) Is π rational?

(ii) Compute $4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7}\right)$ and compare with 3.1416.

(iii) Name the mathematician who proved π irrational and the year.

OR

(i) Show that between any two rational numbers there is another rational number. **5**

(ii) Find the rational number midway between $-\frac{1}{4}$ and $\frac{5}{12}$ and place it on the number line (describe).

END OF QUESTION PAPER **Total: 80 Marks**

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